Utkarsh Bharadwaj

Official Mail | LinkedIn | GitHub | Portfolio

About Me

Mathematics and statistics student at the Indian Statistical Institute, Delhi, passionate about applying probabilistic and optimization frameworks to real-world problems. Eager to contribute to quantitative research and data-driven roles.

Education

Indian Statistical Institute — **Delhi, India**
Bachelor of Statistical Data Science (BSDS) — *Aug 2025 – Present*

- Aggregate: **70.4%** (1st Semester); **84%** (2nd Semester)
- **Coursework**: R Programming, Linear Algebra, Probability Theory, Real Analysis, Statistical Inference, Hypothesis Testing, Linear Regression, Sampling Theory, Optimization Theory and Numerical Methods

Technical Skills

Languages: Python, R, LaTeX

Libraries & Frameworks: Pandas, NumPy, ggplot2, lattice

Statistical Methods: Bayesian Inference, Poisson Processes, QP/Convex Optimization, Benford's Law, Regression Analysis, Hypothesis Testing, Interval Estimation

Tools & Environment: Linux (Debian), VS Code, RStudio, GitHub

Projects

Neyman-Pearson Lab: Hypothesis Testing Tool | *Hypothesis Testing, MP Test* — *July 2026*

- Built a single-page interactive web application that derives Most Powerful (Neyman-Pearson), UMP (Karlin-Rubin), and UMPU level-α tests live for nine one-parameter distribution families.
- Designed a unified derivation engine producing closed-form critical regions and step-by-step, KaTeX-rendered proofs of the rejection region for user-specified hypotheses and significance levels.

CipherInk: Forensic Stylometry Platform | *Bayesian Inference, Statistical NLP* — *June 2026*

- Developed a full-stack, multi-author forensic stylometry web application for authorship attribution.
- Engineered a backend using Poisson log-likelihood and Bayesian softmax to compare statistical fingerprints of 70 filler words against a calibrated global English baseline.

Beyond the Deficit: India's Trade Dynamics | *R, Time-Series Analysis, Macroeconomics* — *May 2026*

- Conducted a 35-year empirical time-series analysis (1990–2025) of India's macroeconomic trade trends.

- Evaluated structural impacts of currency depreciation and energy dependency using inflation-adjusted (real) values.

What Makes a Hit? Billboard Analytics 2018–2025 | *Python, R, API Integration* *May 2026*

- Investigated acoustic and structural properties of Billboard Top 100 songs across eight years.
- Engineered a unified dataset bridging Spotify, Maven Analytics, and AcousticBrainz APIs; applied KNN imputation (Python) and statistical analysis (R).

Convex Optimization for LPG Allocation | *QP, Optimization Theory* *Apr 2026 – May 2026*

- Formulated a strictly convex Quadratic Programming (QP) model to optimally allocate constrained national fuel supply across domestic and commercial sectors.
- Designed an objective function minimizing the weighted sum of squared shortfalls to balance commercial economic losses against environmental health risks, using historical consumption data and demographic vulnerability weights.

Eco-Aware AI Tool (Team Diazonium) | *Python, ASTs, Statistical Modeling* *Mar 2026*

- **Top 10 of 150+ teams** at NIT Delhi UpVision Hackathon ("AI for Humanity" track); built an AST-based tool to estimate carbon/water footprint of Python code in 12 hours, and formulated the statistical validation methodology.

Statistical Critique of Climate Economics | *R, Critical Analysis* *Sept 2025*

- Analyzed impact of GDP data anomalies (e.g., Uzbekistan) on global climate commitment models; critiqued methodological robustness of *Kotz et al. (2024)* vs. *Hsiang et al. (2025)* via sensitivity analysis of distributed lag models.

Certifications & Workshops

Indian Institute of Technology Ropar **Ropar, India**
NS2ML-DE 2026 Workshop *Jul 6 – 10, 2026*

- Covered numerical PDE/ODE solvers (finite difference/element methods) and data-driven ML techniques (Physics-Informed Neural Networks), with hands-on MATLAB/Python labs.

Achievements

ISI Merit List: Secured All India Rank **31** for the BSDS program at the Indian Statistical Institute.

JEE Main 2025: Secured **99.37 percentile** in Mathematics, placing in the top 0.6% of applicants nationwide.

Vihaan 9.0 Hackathon Finalist: Advanced to the offline finals at Delhi Technological University with Team Diazonium.

NIT Delhi UpVision Hackathon: Placed Top 10 of 150+ teams in the "AI for Humanity" track with Team Diazonium.

Languages

Maithili (Mother Tongue), Hindi (Native Speaker), English (Advanced – IELTS Certified, C1 Level).